Tom Hess

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Product-Focused, AI-Enabled Engineering Leader & Software Engineer

Software engineer and engineering leader with 15+ years of experience building automation, integrations, and AI-enabled systems for SaaS and enterprise businesses, including two Director-level engineering roles. Hands-on work includes Java gift card and call-routing systems and RPA-driven billing automation. Since 2025, I have shipped apps and built an agent pipeline using an AI coding agent. Three years in product roles added a product lens, so I prioritize engineering work by customer outcome.

CORE SKILLS

Agentic AI Systems Design & Orchestration • AI-Assisted Full-Stack Application Development (Python, TypeScript, Rust) • AI-Assisted Delivery & Workflow Automation • API / ERP / EDI Integrations • SaaS Platform Modernization • Process Automation & Workflow Optimization • Root Cause Analysis • Technical Debt Management • SQL & Data Analysis • Team Development & Coaching • Cross-Functional Leadership

SELECTED ENGINEERING PROJECTS

Independent Software Projects | Personal AI & Application Development **2025 – Present**

(Personal projects, built with an AI-enabled coding agent • Crosscue • CableHop • Retirement Planner)

- Built a 7-stage, local-first LLM agent pipeline with phase gates, then retired it for a leaner hosted harness when usage data showed guarded review wasn't worth maintaining.
- Shipped Crosscue, a Flutter crossword app, to the iOS App Store and Google Play, with automated daily puzzle retrieval and an account-free leaderboard on Cloudflare Workers and D1.
- Shipped CableHop to the Mac App Store, a Swift macOS app with a custom C USB/MTP engine, for managing Android devices over MTP.
- Rewrote a vanilla JS retirement planner as a React app, running the Monte Carlo simulation in a Web Worker off the main thread, with Zod-validated forms and reducer-based state.

EXPERIENCE

Raptor Technologies | Senior Product Owner **Oct 2023 – Oct 2026**

(Acquired PublicSchoolWORKS, May 2024; Senior Product Manager until Oct 2025, then Senior Product Owner)

- Established team AI governance: approved AI tools and MCP servers, AI-specific logins for data protection, and a merge gate pairing automated AI review with a final human check on all AI-generated code.
- Led product development for compliance-driven LMS and workflow automation platforms serving ~45,000 schools nationwide, with ~99.9% customer retention.
- Eliminated regressions within 90 days through structured peer review, QA pipelines, and domain ownership, reducing support tickets by 20–30% and generating unsolicited positive customer feedback.
- Oversaw two delivery teams post-acquisition: an internal team of 3 engineers, 3 QA, and 1 DevOps lead, plus a vendor partner team of 1 PM, 3 engineers, and shared QA.

Octus (formerly Reorg) | Director of Engineering **Jul 2022 – Sep 2023**

- Led root cause analysis on database and scalability outages, meeting 24–48 hour resolution SLAs on major incidents, where SLAs had previously been rarely met.
- Directed three cross-functional delivery teams with $500K budget oversight: 10 direct engineering reports plus two additional teams managed through an engineering manager.
- Implemented Scrum practices and documentation standards (runbooks, design docs, onboarding guides), reducing engineer onboarding time by 50%.

Cengage Group | Manager, Software Engineering **Jan 2020 – Jul 2022**

- Delivered a RESTful Authorization & Permissions API (first touchpoint for every login) with sub-millisecond latency and 99.9% uptime by consolidating 5 legacy services.

- Retired 3 legacy applications through an SOA modernization project, collaborating with 20+ system owners to reduce complexity, maintenance burden, and costs.
- Increased sprint velocity by 50% through SAFe Agile practices, cross-team alignment, and coaching engineers into expanded roles.

Phillips Edison & Company | Director / Associate VP, Enterprise Applications **Oct 2013 – Jan 2020**

(Progressed from Application Developer → Manager → Director → Associate VP over 6+ years)

- Built Tenant Billing Automation Platform integrating ERP, vendor APIs (Billtrust), and RPA, cutting a 2-week billing cycle by 50–60% and freeing billing staff for direct tenant service and other higher-value work.
- Developed the Lease Profitability App alongside Finance and Leasing agents, cutting deal analysis from hours to seconds and giving agents real-time deal evaluation.
- Grew team from two developers to 12 engineers, contractors, and interns, training them on MRI, a real estate ERP, while overseeing a ~$1M annual budget.

Additional Technical Experience **Sep 2005 – Oct 2013**

(The Ohio State University • KeyBank • Barnes Distribution • Sterling Jewelers)

- Built a nationwide gift card processing application in Java for Sterling Jewelers' 13 brands, eliminating external vendor reliance and bringing processing fully in-house.
- Built an automated IVR/call tree system in Java with real-time API integrations at Sterling Jewelers, reducing average customer call handling time by 25%.
- Led accounts payable side of Barnes Distribution's EDI payments project, automating JD Edwards data-file download and import steps and cutting check printing from 2 days to a couple of hours.

TECHNICAL COACHING & APPLIED EXPERIENCE

FIRST Tech Challenge (FTC) | Robotics Head Coach – Software Development **2023 – Present**

- Coached teams to two World Championship qualifications (2023–24, 2024–25), including a top-4 finish at the Ohio State Championship and the software-specific FTC Control Award in 2024–25.

TEAM & LEADERSHIP RECOGNITION

Tech Team of the Year – Cincinnati Business Courier (2019) • MRI Impact Award – Financial Modeling & Workflow Innovation (2018) • Corporate Tech & Innovation Award – Enterprise Automation (2017) • Realcomm Digie Award – Automation Leadership (2017)

EDUCATION

M.S., Computer Science & Engineering **2007**
The Ohio State University
Thesis: Simulation of a Hybrid Vehicle Based on Software Components

B.S., Computer Science & Engineering **2003**
The Ohio State University

CERTIFICATIONS

ITIL Foundation Certificate • Certified BMC Remedy Administrator • Claude Code 101 • Google Cloud Agent Fundamentals

COMMUNITY & VOLUNTEERING

Crayons to Computers – Supply Distribution for Underserved Students • Hamilton County Parks – Seasonal Cleanup & Restoration Volunteer